

# AI Student Learning Path - Project Report

## 1. Project Overview

A comprehensive AI-driven system that analyzes student learning patterns using Artificial Neural Networks (ANN) and provides personalized recommendations.

## 2. Project File Structure

```
AI_Student_Learning_Path/
|-- dataset/           # Dataset and training scripts
|-- model/             # Trained AI models and scalers
|-- static/            # Static assets for Flask
|-- templates/         # HTML templates for Flask UI
|-- uploads/           # Student marksheet uploads
|-- flask_app.py        # Main Flask backend
|-- show_db.py         # Database verification
|-- requirements.txt    # Library dependencies
|-- README.md          # Overview documentation
|__ project_report.md   # Detailed report
```

## 3. Entity Relationship Diagram (ERD)

The database consists of four primary entities: USERS, ASSESSMENTS, COURSES, and QUESTIONS. Users can take multiple assessments, creating a one-to-many relationship.

```
USERS (id, email, password, name, roll_number, interest, current_level)
|
|--< ASSESSMENTS (id, user_id, subject, marks, predicted_level, confidence, next_steps)

COURSES (id, level, title, url, description)
QUESTIONS (id, subject, question_text, correct_option)
```

## 4. Data Flow Diagram (DFD Level 0)

The Context Diagram shows the Student interacting with the AI Learning Path System by providing assessment data and receiving predictions.

## 5. Data Flow Diagram (DFD Level 1)

- P1: Authentication - Validates user identity.

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- P2: Assessment Collection - Captures marks and study time.
- P3: AI Level Prediction - Neural Network processes metrics.
- P4: Recommendation Generation - Maps level to resources.
- P5: Progress Visualization - Renders performance charts.

## 6. Technical Stack

- Backend: Flask & MySQL
- AI/ML: TensorFlow (ANN), Scikit-Learn
- Frontend: Jinja2 Templates & CSS
- Analytics: Matplotlib & Seaborn